Ans : (b) India's foreign exchange reserves consist of the foreign-currency assets, gold, special drawing rights and the reserve tranche position in the International Monetary Fund. It does not consist of the loans from World Bank or from foreign countries.

66. The balance of payments of a country is a systematic record of –
(a) all import and export transactions of a country during a given period of time, normally a year
(b) goods exported from a country during a year
(c) economic transaction between the government of one country to another
(d) capital movements from one country to another

Ans : (a) The Balance of payment (BoP) record the transactions in goods, services and assets between residents of a country with the rest of the world for a specified time period, typically a year. So the best option is A.

SCIENCE

• Physics

67. Ball bearings are used in bicycles, cars, etc. because –
(a) The actual area of contact between the wheel and axle is increased
(b) The effective area of contact between the wheel and axle is increased
(c) The effective area of contact between the wheel and axle is reduced
(d) None of the above statements is correct

Ans : (c) The roundness of the bearing creates less contact between the bearing that is trying to move objects and the objects being moved. A reduction in contact area means less friction.

68. Consider the following :
1. Electromagnetic radiation
2. Geothermal energy
3. Gravitational forces
4. Plate movements
5. Rotation of the earth
6. Revolution of the earth

Which of the above are responsible for bringing dynamic changes on the surface of the earth ?
(a) 1, 2,3, and 4 (b) 1,3,5 and 6 only
(c) 2,4,5 and 6 only (d) 1, 2, 3, 4, 5 and 6

Ans : (d) Endogenous forces have a role in bringing dynamic changes to the surface of the earth. Endogenous forces are generated due to thermal asymmetry in the interior of the Earth and contraction and expansion of rocks. Electromagnetic radiation, the gravitational force also has a role. The rotation, revolution and geothermal energy of the Earth is also a reason to bring in dynamic changes on the Earth's surface.

69. Consider the following phenomena :
1. Size of the sun at dusk
2. Colour of the sun at dawn
3. Moon being visible at dawn
4. Twinkle of stars in the sky
5. Polestar being visible in the sky

Which of the above are optical illusions ?
(a) 1, 2 and 3 (b) 3, 4 and 5
(c) 1, 2 and 4 (d) 2, 3 and 5

Ans : (c) The following are the optical illusions-
- Shimmer of the stars.
- The red appearance of the sun at sunrise and sunet, it happen due to the refraction of light.
- While, polestar and moon are visible at dawn in the actual (real) scenario.

70. Rainbow is produced when sunlight falls on drops of rain. Which of the following physical phenomena are responsible for this ?
1. Dispersion
2. Refraction
3. Internal reflection

Select the correct answer using the codes given below
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 (d) 1,2 and 3

Ans : (d) Option–1, Each individual droplet of water acts as a tiny prism that both disperses the light and reflects it back to our eye.
Option–2, Since water is more denser than air, lights is refracted as it enters the drop.
Option–3, Some of the light will reflect off the back of the drop if the angle is larger than the critical angle (48º for water).

71. The known forces of nature can be divided into four classes, viz., gravity, electromagnetism, weak nuclear force and strong nuclear force. With reference to them, which one of the following statements is not correct ?
(a) Gravity is the strongest of the four
(b) Electromagnetism acts only on particles with an electric charge
(c) Weak nuclear force causes radioactivity
(d) Strong nuclear force holds protons and neutrons inside the nucleus of an atom

Ans : (a) The known forces of nature are divided into four classes. These forces are gravity force, electromagnetism force, weak nuclear force and strong nuclear force. The decreasing order of these forces is as follows-strong nuclear force > electromagnetism force > weak nuclear force > Gravitational force.

72. The efforts to detect the existence of Higgs boson particle have become frequent news in the recent past. What is / are the importance/ importances of discovering this particle ?
1. It will enable us to understand as to why elementary particles have mass.
2. It will enable us in the near future to develop the technology of transferring matter from one point to another without traversing the physical space between them.
3. It will enable us to create better fuels for nuclear fission.

Select the correct answer using the codes given below.
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (a) Higgs Boson is a particle postulated by the standard model of physics to provide mass to all particles in the universe. Discovery of the Higgs Boson will help us to further understand why elementary particles have mass.

73. During a thunderstorm, the thunder in the skies is produced by the
1. meeting of cumulonimbus clouds in the sky
2. lightning that separates the nimbus clouds
3. violent upward movement of air and water particles.

Select the correct answer using the codes given below :
(a) 1 only
(b) 2 and 3
(c) 1 and 3
(d) None of the above produces the thunder

Ans : (d) Thunderstorm is an intense local thunderstorm, with wide and dense rainy cumulus clouds and a strong wind moving from the bottom to the top. A thunderstorm often arises due to atmospheric instability, i.e. low pressure and intense convective airflow. In fact, thunderstorms are local storms in which the winds blow upward at high speeds, accompanied by lightning and thunder clouds and also heavy rain.

• Medical Science (Zoology)

74. Which of the following statements is/are correct ?
1. Viruses lack enzymes necessary for the generation of energy
2. Viruses can be cultured in any synthetic medium
3 Viruses are transmitted from one organism to another by biological vectors only.

Select the correct answer using the codes given below :
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (a) Statement 1 is correct and viruses do not have the capacity to survive on their own and they need a living host for survival. Statement 2 is incorrect; viruses cannot grow in a synthetic medium. Statement 3 is incorrect because viruses enter the body from the environment or other individuals from soil to water or air via nose, mouth, or any cracks in the skin and seek a cell to infect.

75. Which of the following diseases can be transmitted from one person to another through tattooing?
1. Chikungunya
2. Hepatitis B
3. HIV-AIDS

Select the correct answer using the codes given below :
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (b) AIDS and hepatitis B are diseases that are transmitted from person to person through tattooing or infection. On the other hand, chikungunya is a viral disease and is spread by mosquito bites. Its carrier is mosquito.

76. Consider the following minerals :
1. Calcium
2. Iron
3. Sodium

Which of the minerals given above is /are required by human body for the contraction of muscles ?
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d) Calcium : Calcium is essential for activating enzymes that cause muscle contraction.
Iron : Iron helps support continued muscle function by supporting energy production in your muscle cells so that muscles fibres always have the energy they need to contract properly. Iron also helps muscles store oxygen to fuel muscle contractions & promotes healthy circulation so that muscles can get additional oxygen from the bloodstream.
Sodium : One need sodium for muscle contraction because it balances potassium to maintain membrane potential.

77. Recombinant DNA technology (Genetic Engineering) allows genes to be transferred
1. across different species of plants
2. from animals to plants
3. from microorganisms to higher organisms

Select the correct answer using the codes given below
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d) Recombinant DNA are DNA sequences, which result from bringing genetic material from different sources. Here the genes can be transferred between any species, that is, across different species of plants from animals to plants and for microorganisms to higher organisms. Thus, all three options are correct.

78. Consider the following animals :
1. Sea cow
2. Sea horse
3. Sea lion

Which of the above is/are mammal/ mammals ?
(a) 1 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3